

Are Concepts Independent? Hierarchy-Aware Concept Activation Vectors

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Abstract

Concept Activation Vectors (CAVs) are a widely used tool for interpreting neural network representations, yet standard training ignores the hierarchical structure of real data: when a child concept usually occurs within a parent context (e.g., camping implies an open area), treating the two as independent yields misleading latent-space comparisons. We distinguish empirically-discovered hierarchies (from co-occurrence) from semantic ones (from language-model entailment), and evaluate five CAV strategies across 12 hierarchies on the SUN Attribute Database: a mean-difference baseline, two orthogonalisation variants, our parent-conditioned Filtering method, and a supervised probe under both standard and conditional (parent-present) AUC, on ResNet18-Places365 and CLIP ViT-B/32. Filtering is the only unsupervised variant with a clear benefit, and only for empirically-grounded hierarchies: it improves conditional AUC by ~ 0.5 pp on empirical pairs but reduces it on semantic ones by 0.6–1.6pp, consistently across both backbones. This suggests data-driven filtering helps only when hierarchical structure is reflected in dataset co-occurrence, not when it is purely semantic.

1. Introduction

Concept Activation Vectors (CAVs) are a standard tool for interpreting deep vision models, using a linear direction in activation space to separate a concept from counter-examples Kim et al. (2018). This rests on the assumption that concepts are encoded as linear directions, a framework formalized for language models by Park et al. (2024). Standard CAVs implicitly assume concept independence, yet real visual concepts are often hierarchical: in the SUN Attribute Database, *carpet* co-occurs with *enclosed area* in 96% of images, and natural-language-inference models judge that *ice* entails *cold* with 0.99 probability. When a child concept usually appears within a parent context, the negative set used to train its CAV is dominated by out-of-context images, so the learned direction conflates the child with its parent (Figure 1A). This inflates separability on the full test set while degrading discrimination exactly where it matters among images sharing the parent context. Our core idea is to make CAV construction *parent-aware* (Figure 1B). Instead of contrasting the child against all counter-examples, our Filtering CAV restricts both the positive and negative sets to images where the parent is present, and takes the difference of their centroids. The resulting direction points from the parent-context centroid toward the child, isolating what distinguishes the child *within* its parent context rather than the parent context it-

self. This targets the regime where standard CAVs fail, and motivates a complementary *conditional* AUC metric that scores probes only on parent-present images. This framing raises a question prior work has not examined: does the hierarchy’s *origin* matter? Hierarchies can emerge empirically (from dataset co-occurrence) or semantically, and these sources often diverge. We study how parent-aware CAVs behave across both origins, and how this varies between a purely visual (ResNet-18-Places365) and a language-supervised (CLIP ViT-B/32) backbone.

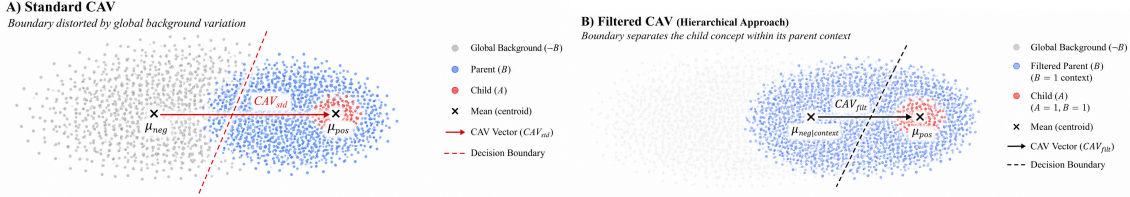


Figure 1: Illustration of the Filtering CAV. By intersecting the parent and child sets, computing the centroid of each, and constructing a vector pointing from the parent centroid toward the child, the algorithm separates the child concept (red dots) from its parent (blue dots) within the parent context, in contrast to the standard mean-difference CAV.

To address this, we make the following contributions:

1. We propose two parent-aware CAV variants Filtering and HCEP-Filtering and a conditional AUC metric that evaluates concept probes only on parent-present images, alongside an orthogonalisation baseline (HCEP) Nguyen et al. (2026).
2. We identify 12 concept hierarchies in the SUN Attribute Database: six from co-occurrence statistics ($P(B | A) \geq 0.87$) and six from NLI entailment, evaluated on ResNet18-Places365 and CLIP ViT-B/32.
3. We show that Filtering improves conditional AUC over the standard mean-difference CAV by ~ 0.5 pp on empirical hierarchies but reduces it by 0.6–1.6pp on semantic hierarchies, consistently across both backbones (Tables 6–7).

2. Related Work

Recent work on hierarchical vision-language embeddings primarily follows two approaches. The HCEP framework proposed by Nguyen et al. (2026) imposes explicit geometric constraints to achieve strong few-shot recovery, but requires a predefined hierarchy and large embedding dimensions. Alternatively, Costa et al. (2025) use an unsupervised sequential autoencoder to discover *conditional orthogonality*, though its greedy inference can struggle under strong feature entanglement. While both frameworks leverage the insight that child-parent difference vectors are near-orthogonal to their parents, neither applies it to correct a *linear concept probe*. Our Filtering method addresses this gap by constructing hierarchy-aware CAVs, restricting the negative training set to images sharing the parent context. We evaluate this correction across both empirically and semantically discovered structures.

3. Methodology

3.1 Selection of dataset, concept pairs and models

We use the SUN Attributes Database (Patterson and Hays, 2012), comprising 14,340 scene images with 102 binary attributes. This dataset was selected because its multi-label nature naturally include complex real-world concept hierarchies, making it ideal for studying contextual entanglement. From this dataset, we define 12 hierarchical pairs (A, B) , where A is the child concept and B is the parent. These pairs are categorised by their discovery mechanism. The first category comprises 6 *empirical pairs* extracted via dataset statistics, exhibiting strong conditional co-occurrence $P(B|A) \geq 0.87$ (e.g., *carpet* $\rightarrow$ *enclosed area*). The second category contains 6 *semantic pairs* derived using language entailment. Specifically, we scored potential concept relationships using the **cross-encoder/nli-deberta-v3-small** (He et al., 2023) natural language inference model, and then manually selected the 6 most contextually sound hierarchies (e.g., *conducting business* $\rightarrow$ *work*). Some of them also have strong conditional co-occurrence. Full details of these pairs are provided in Appendix B. To evaluate whether hierarchy handling depends on the nature of the latent space, we compare two architecturally distinct models. We use ResNet18-Places365 (Zhou et al., 2018) as a purely visual backbone supervised on scene classification, extracting its 512-dimensional penultimate-layer activations. Conversely, we use CLIP ViT-B/32 (Radford et al., 2021) to provide a multimodal latent space, utilising its 512-dimensional L2-normalised image embeddings.

3.2 CAV Construction Methods

Given a dataset subset, we construct a child CAV $\mathbf{v}_A$ to isolate the concept A using five distinct approaches. The standard baseline, MEANDIFF, computes the mean-difference vector $\mathbf{v}_A \propto \mu_A^+ - \mu_A^-$ over the entire training set. To explicitly address hierarchy, HCEP Nguyen et al. (2026) orthogonalises the baseline child vector against the parent vector such that $\mathbf{v}_A^{\text{HCEP}} \propto \mathbf{v}_A - (\mathbf{v}_A \cdot \mathbf{v}_B)\mathbf{v}_B$, guaranteeing $\mathbf{v}_A^{\text{HCEP}} \perp \mathbf{v}_B$. Our proposed method, FILTERING (illustrated in Figure 2), tackles this entanglement directly at the data level. Instead of post-hoc geometric correction, it restricts the positive and negative sets strictly to images where the parent context is present, computing $\mathbf{v}_A^{\text{FILTERING}} \propto \mu_{A|B}^+ - \mu_{A|B}^-$. Finally, we evaluate HCEP-F, a hybrid approach that deploys a scoring vector $\mathbf{v}_A - \mathbf{v}_B$ alongside a hierarchical inference gate. These are additionally compared against a standard Logistic Regression classifier trained to predict A from the unmodified embeddings.

3.3 Inference and Evaluation Protocol

To detect a concept from a computed CAV v , we threshold the cosine similarity: $\hat{y}(x) = \mathbb{1}[s(x, v) \geq \tau^*]$, where the per-concept threshold τ^* is chosen via an on-the-fly grid search maximising training-set F1 (avoiding a hardcoded symmetric threshold or Gaussian assumptions on the latents; values in Appendix A). For HCEP-F the gate is applied hierarchically: a child is rejected if the parent score fails τ_B^* . We evaluate via ROC-AUC on a fixed 80/20 train/test split under two regimes: *Standard AUC* on the full test set, and *Conditional AUC*, restricted to hard negatives test samples where the parent concept is present ($B = 1$).

4. Experiments and Results

Table 1 summarizes mean rank of methods across the 12 hierarchies for all five CAV variants MeanDiff (MD), HCEP, HCEP-Filtering (HCEP-F), Filtering (Filt), and the supervised logistic-regression baseline (CAV) under Standard and Conditional AUC, with critical-difference diagrams in Figure 3 (Appendix E) and full per-pair values in Tables 4–7. Under Standard AUC, the unmodified CAV baseline ranks best on both models (1.75 ResNet18, 1.13 CLIP), with MeanDiff the best centroid-based variant. HCEP and HCEP-Filtering consistently underperform MeanDiff, most severely on weakly coupled pairs. Under Conditional AUC, Filtering ranks best on ResNet18 (2.17, ahead of CAV at 2.38), matching or exceeding MeanDiff on 8/12 pairs; on CLIP, CAV remains best (1.08). HCEP and HCEP-Filtering stay weakest in both settings. CAV’s largest advantages occur on *conducting business* (+0.127 ResNet18) and *rusty* (+0.032 ResNet18, +0.045 CLIP) both semantic pairs with weak co-occurrence ($P(B | A) < 0.5$) suggesting that where a hierarchy is not grounded in strong dataset co-occurrence, a fully discriminative probe captures context the centroid-based corrections miss. Full results can be found in Appendix D.

Table 1: Mean AUC rank across the 12 hierarchies for each CAV variant (1 = best, 5 = worst), under Standard AUC and Conditional AUC (parent=1). Values match Figure 3.

Method	Standard AUC		Conditional AUC	
	ResNet18	CLIP	ResNet18	CLIP
MeanDiff	2.50	2.88	2.92	2.96
HCEP	3.46	3.54	3.04	3.58
HCEP-Filtering	4.46	4.38	4.50	4.63
Filtering	2.83	3.08	2.17	2.75
CAV	1.75	1.13	2.38	1.08

5. Conclusions

We investigated whether imposing hierarchical structure on CAV construction by orthogonalisation (HCEP, HCEP-F) or parent-conditioned training (FILTERING) improves concept detection, on 12 hierarchies drawn from empirical co-occurrence and semantic entailment. Filtering is the only hierarchical variant with a clear benefit, and only for pairs with strong empirical co-occurrence: it ranks best under Conditional AUC on ResNet18 and matches or exceeds MeanDiff on most such pairs. For semantically-motivated, weakly-coupled pairs ($P(B | A) < 0.5$) it does not help, and the supervised baseline retains a large advantage. HCEP and HCEP-Filtering are consistently the weakest methods, suggesting that projecting out the parent direction removes more useful signal than parent-context leakage. These conclusions are limited by the small evaluation set (12 hierarchies, one split), which leaves several comparisons statistically inconclusive; scaling to more hierarchies and multiple seeds is the key next step.

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Appendix A. Pre-calculated optimal thresholds

Table 2 presents the pre-calculated thresholds used to evaluate the cosine similarity between a given test vector x and a parent concept. Establishing whether an observation belongs to the parent concept is a crucial gating step; it confirms the hierarchical context and determines whether the subsequent child CAV evaluation should be applied. These thresholds were empirically selected by performing a grid search on the training set to maximize the F_1 score. During the test phase, this pre-computed dictionary can be applied directly to ensure efficient inference.

Table 2: Pre-calculated optimal thresholds τ_B^* for parent concepts, derived from F1 grid search on the CLIP training features.

Parent concept B	τ_B^*	Best F1
enclosed area	0.067	0.866
open area	0.000	0.812
natural light	-0.044	0.829
cold	0.300	0.685
asphalt	0.178	0.521
working	0.156	0.565
rock/stone	0.333	0.488
aged/worn	0.222	0.375

Appendix B. Hierarchies found in a dataset

Table 3: Hierarchies found in the SUN Attribute Dataset. Empirical hierarchies were identified via co-occurrence statistics ($P(B | A) \geq 0.87$); semantic hierarchies were identified via NLI entailment scoring and are included regardless of their co-occurrence statistic. That is the reason they may have relatively high $P(B | A)$. Types describe only the method how we found them.

Child (A)	Parent (B)	$P(B A)$	n_A	n_B	Type
carpet	enclosed area	0.96	324	4947	Empirical
research	enclosed area	0.95	207	4947	Empirical
medical activity	enclosed area	0.92	118	4947	Empirical
camping	open area	0.89	871	5541	Empirical
gaming	enclosed area	0.88	108	4947	Empirical
sand	natural light	0.87	351	6709	Empirical
ice	cold	0.88	158	388	Semantic
ocean	open area	0.85	413	5541	Semantic
conducting business	working	0.40	263	1032	Semantic
climbing	rock/stone	0.36	638	594	Semantic
pavement	asphalt	0.33	632	635	Semantic
rusty	aged/ worn	0.29	280	631	Semantic

Appendix C. Filtering algorithm

Training phase

A - child concept, B - parent concept

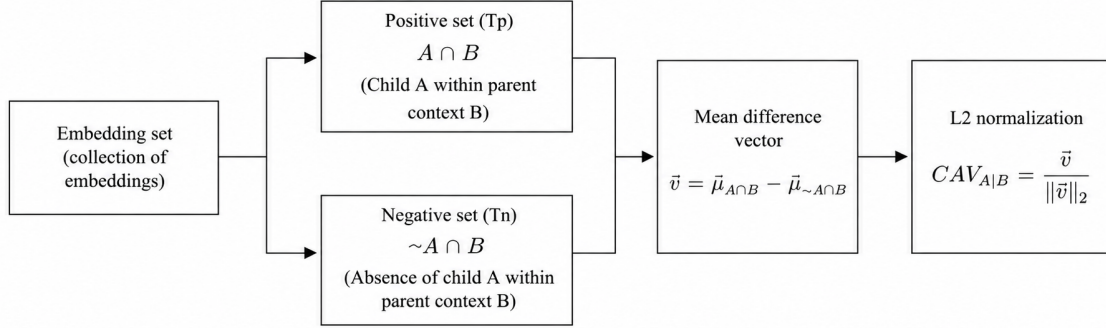


Figure 2: Filtering isolates the child concept (e.g., *asphalt*) by restricting the optimisation space strictly to the parent domain (*pavement*), explicitly disentangling their features.

Appendix D. Full AUC Tables

Notes on Table 4–7. MD denotes the standard mean-difference CAV, HCEP and HCEP-F (HCEP-Filtering) denote the orthogonality-based variants, Filt denotes the context-restricted Filtering CAV, and CAV denotes the fully supervised logistic-regression baseline. Tables 4 and 5 report Standard AUC (full test set, gated by the parent threshold); Tables 6 and 7 report Conditional AUC, restricted to test images where the parent attribute is present ($B = 1$), with the best-performing method per pair shown in bold.

Table 4: Standard AUC — ResNet18-Places365.

Child	Parent	MD	HCEP	HCEP-F	Filt	CAV
research	enclosed area	0.958	0.922	0.941	0.944	0.943
camping	open area	0.959	0.916	0.949	0.959	0.959
medical activity	enclosed area	0.980	0.960	0.968	0.975	0.984
gaming	enclosed area	0.965	0.948	0.956	0.958	0.909
sand	natural light	0.955	0.953	0.898	0.957	0.963
carpet	enclosed area	0.914	0.789	0.877	0.832	0.912
ocean	open area	0.982	0.981	0.868	0.982	0.980
conducting business	working	0.893	0.802	0.780	0.789	0.879
pavement	asphalt	0.898	0.735	0.799	0.781	0.903
climbing	rock/stone	0.916	0.760	0.844	0.857	0.947
rusty	aged/ worn	0.898	0.828	0.844	0.820	0.882
ice	cold	0.982	0.757	0.936	0.915	0.992

Table 5: Standard AUC — CLIP ViT-B/32.

Child	Parent	MD	HCEP	HCEP-F	Filt	CAV
research	enclosed area	0.967	0.928	0.941	0.944	0.974
camping	open area	0.946	0.870	0.913	0.948	0.968
medical activity	enclosed area	0.990	0.979	0.976	0.983	0.990
gaming	enclosed area	0.976	0.964	0.959	0.969	0.983
sand	natural light	0.937	0.931	0.916	0.936	0.978
carpet	enclosed area	0.902	0.785	0.864	0.841	0.934
ocean	open area	0.979	0.964	0.952	0.974	0.986
conducting business	working	0.918	0.828	0.842	0.796	0.931
pavement	asphalt	0.883	0.729	0.802	0.748	0.911
climbing	rock/stone	0.908	0.781	0.815	0.864	0.951
rusty	aged/ worn	0.898	0.845	0.844	0.811	0.934
ice	cold	0.984	0.771	0.886	0.937	0.997

Table 6: Conditionall AUC (parent=1) — ResNet18-Places365. Best method per pair in bold.

Child	Parent	MD	HCEP	HCEP-F	Filt	CAV
research	enclosed area	0.888	0.896	0.889	0.895	0.872
camping	open area	0.911	0.903	0.892	0.916	0.906
medical activity	enclosed area	0.946	0.948	0.939	0.950	0.961
gaming	enclosed area	0.947	0.945	0.929	0.949	0.881
sand	natural light	0.916	0.934	0.845	0.927	0.935
carpet	enclosed area	0.770	0.769	0.755	0.773	0.796
ocean	open area	0.967	0.971	0.867	0.972	0.967
conducting business	working	0.646	0.693	0.676	0.624	0.773
pavement	asphalt	0.619	0.579	0.569	0.633	0.632
climbing	rock/stone	0.864	0.822	0.807	0.868	0.856
rusty	aged/ worn	0.767	0.767	0.758	0.750	0.799
ice	cold	0.849	0.796	0.814	0.829	0.837

Table 7: Conditionall AUC (parent=1) — CLIP ViT-B/32. Best method per pair in bold.

Child	Parent	MD	HCEP	HCEP-F	Filt	CAV
research	enclosed area	0.918	0.910	0.892	0.914	0.934
camping	open area	0.880	0.860	0.835	0.894	0.924
medical activity	enclosed area	0.974	0.969	0.954	0.970	0.974
gaming	enclosed area	0.951	0.959	0.950	0.959	0.965
sand	natural light	0.886	0.910	0.869	0.902	0.956
carpet	enclosed area	0.760	0.761	0.751	0.763	0.827
ocean	open area	0.960	0.958	0.933	0.963	0.975
conducting business	working	0.775	0.720	0.717	0.696	0.821
pavement	asphalt	0.600	0.568	0.588	0.597	0.625
climbing	rock/stone	0.838	0.818	0.789	0.844	0.844
rusty	aged/ worn	0.787	0.780	0.802	0.772	0.832
ice	cold	0.857	0.813	0.828	0.852	0.887

Appendix E. Critical difference diagrams

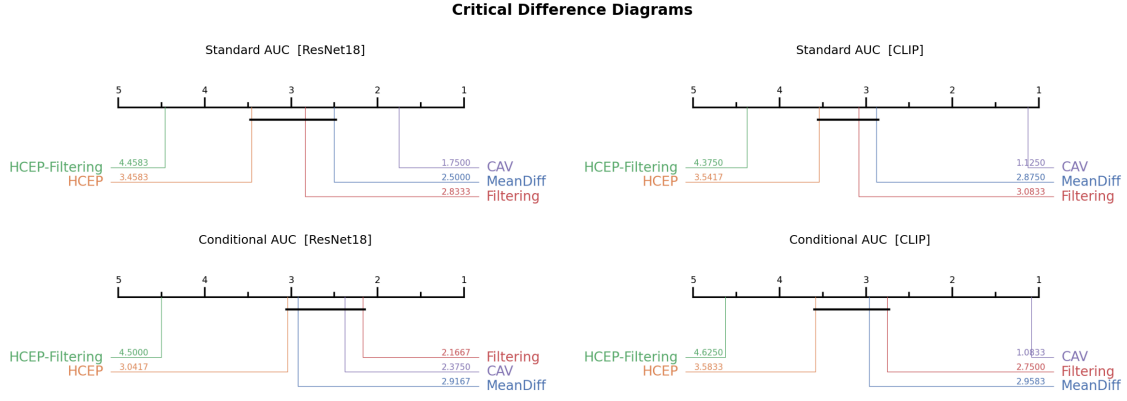


Figure 3: Critical difference diagrams (Wilcoxon signed-rank test, Holm correction, $\alpha = 0.1$) across the 12 hierarchies, for Standard AUC (top) and Conditional AUC (bottom), on ResNet18-Places365 (left) and CLIP ViT-B/32 (right). Thick horizontal bars connect methods with no statistically significant difference. Numbers indicate mean rank (Table 1).